

ePG Introduction to Machine Learning Tutorial

Convexity

Prof Jayakrishnan and Prof. Sharayu Mahavir
TA : Pinak Mahapatra

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Instructions

- Show all derivations clearly.
- Justify all theoretical arguments.
- Use geometric intuition wherever applicable.
- Unless otherwise stated, assume all vectors belong to $\mathbb{R}^n$.

Question 1: Convex Combinations

Let $x, y \in \mathbb{R}^n$ and $\alpha \in [0, 1]$. A convex combination of x and y is given by

$$z = \alpha x + (1 - \alpha)y.$$

- Explain geometrically why z lies on the line segment joining x and y .
- What happens when $\alpha = 0$ and $\alpha = 1$?
- What happens if $\alpha < 0$ or $\alpha > 1$? Explain why the resulting point need not lie on the line segment joining x and y .

Solution

- We can rewrite the convex combination as

$$z = \alpha x + (1 - \alpha)y = y + \alpha(x - y).$$

Since $\alpha \in [0, 1]$, as α varies from 0 to 1, z moves continuously from y to x .

Therefore, z lies on the line segment joining x and y .

(b) If $\alpha = 0$, then

$$z = 0x + (1 - 0)y = y.$$

Thus, $z = y$.

If $\alpha = 1$, then

$$z = 1x + (1 - 1)y = x.$$

Thus, $z = x$.

Hence, $\alpha = 0$ and $\alpha = 1$ correspond to the two endpoints of the line segment.

(c) If $\alpha < 0$ or $\alpha > 1$, the point generally lies outside the line segment.

For example, let $\alpha = 2$. Then

$$z = 2x + (1 - 2)y = 2x - y.$$

This point lies on the line passing through x and y , but outside the line segment joining them.

Therefore, the condition

$$0 \leq \alpha \leq 1$$

is necessary for z to lie on the line segment.

Question 2: Convex Sets

A set $\mathcal{X} \subseteq \mathbb{R}^n$ is called convex if, for every $x, y \in \mathcal{X}$ and every $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in \mathcal{X}.$$

- (a) Explain the geometric meaning of the definition of a convex set.
- (b) Why is a line segment a convex set?
- (c) Give an example of a set that is not convex and explain why it fails the definition.

Solution

- (a) Geometrically, a set is convex if the entire line segment joining any two points in the set is also contained within the set.

Mathematically, for any $x, y \in \mathcal{X}$ and any $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in \mathcal{X}.$$

Therefore, one can choose any two points in the set and move along the straight line between them without leaving the set.

(b) Consider the line segment between two points x and y :

$$\mathcal{S} = \{\alpha x + (1 - \alpha)y : \alpha \in [0, 1]\}.$$

Take any two points $z_1, z_2 \in \mathcal{S}$. Since both points lie on the same line segment, every convex combination

$$\beta z_1 + (1 - \beta)z_2, \quad \beta \in [0, 1],$$

also lies on the same line segment.

Hence, a line segment is a convex set.

1 Question 3: Polyhedral Sets

Consider the set

$$\mathcal{P} = \left\{ x \in \mathbb{R}^2 : \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & -1 \end{bmatrix} x \leq \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix} \right\}.$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Answer the following:

- Write the matrix inequality explicitly as a collection of three scalar linear inequalities in x_1 and x_2 .
- For the first inequality, identify the corresponding boundary line and explain geometrically which side of the line is selected.
- Explain why the set defined by the first inequality is a half-space.

Solution

- We have

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix}.$$

Multiplying the matrices gives

$$\begin{bmatrix} x_1 + 2x_2 \\ -x_1 + x_2 \\ 2x_1 - x_2 \end{bmatrix} \leq \begin{bmatrix} 6 \\ 2 \\ 4 \end{bmatrix}.$$

Therefore, the matrix inequality represents the three scalar inequalities

$$\boxed{x_1 + 2x_2 \leq 6},$$

$$\boxed{-x_1 + x_2 \leq 2},$$

and

$$\boxed{2x_1 - x_2 \leq 4}.$$

Thus, $Ax \leq b$ is simply a compact way of writing multiple linear inequalities.

(b) Consider the first inequality

$$x_1 + 2x_2 \leq 6.$$

Its boundary is obtained by replacing $\leq$ with $=$:

$$x_1 + 2x_2 = 6.$$

This is a straight line in $\mathbb{R}^2$. Rearranging,

$$x_2 = 3 - \frac{x_1}{2}.$$

The inequality

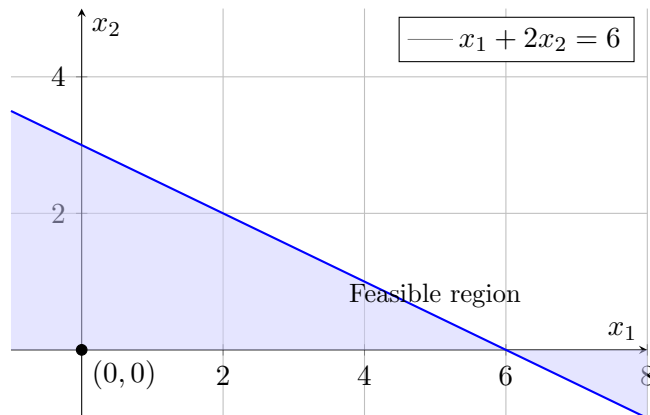
$$x_1 + 2x_2 \leq 6$$

selects all points on or below this line.

For example, the point $(0,0)$ satisfies

$$0 + 2(0) \leq 6,$$

so the side containing the origin is the feasible side.



The shaded region represents all points satisfying $x_1 + 2x_2 \leq 6$. Since $(0,0)$ satisfies the inequality, the side containing the origin is selected.

(c) The line

$$x_1 + 2x_2 = 6$$

divides the plane into two regions:

$$x_1 + 2x_2 \leq 6$$

and

$$x_1 + 2x_2 > 6.$$

The set

$$\{x \in \mathbb{R}^2 : x_1 + 2x_2 \leq 6\}$$

contains one of these two regions together with the boundary line. Such a set is called a half-space.

Hence,

$\{x : x_1 + 2x_2 \leq 6\}$ is a half-space.

Tutorial Problems: Linear Regression & Matrix Formulations

Problem 1: Fitting a Linear Regression Model

Suppose we are given a simple supervised learning dataset with $n = 3$ training samples and $p = 1$ quantitative feature:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 3), \quad (x_3, y_3) = (3, 5).$$

We wish to fit a linear model $f(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ by minimizing the Residual Sum of Squares (RSS).

- (a) Express the dataset in vectorized form by writing down the design matrix $X \in \mathbb{R}^{3 \times 2}$ (including the constant intercept column) and the target vector $Y \in \mathbb{R}^{3 \times 1}$.
- (b) Compute the matrix product $X^T X$ and its inverse $(X^T X)^{-1}$.
- (c) Calculate the optimal coefficient vector $\hat{\beta} = (X^T X)^{-1} X^T Y$ and write down the final prediction function $f(x)$.
- (d) Compute the vector of predicted outputs $\hat{Y} = X \hat{\beta}$ and the residual vector $e = Y - \hat{Y}$.

Solution

- (a) Appending a column of 1s to account for the intercept term, the design matrix X and target vector Y are:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad Y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

- (b) First, we compute $X^T X$:

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

Multiplying the matrices element by element gives:

$$X^T X = \begin{bmatrix} 1+1+1 & 1+2+3 \\ 1+2+3 & 1^2+2^2+3^2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

Next, we find the inverse of $X^T X$. Recall that for any 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

its inverse is given by

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

provided that $\det(A) \neq 0$.

The determinant is calculated as:

$$\det(A) = ad - bc.$$

In our case,

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

Therefore,

$$\det(X^T X) = (3)(14) - (6)(6) = 42 - 36 = 6.$$

Since the determinant is non-zero, the matrix is invertible. Using the 2×2 inverse formula, we swap the diagonal elements (3 and 14), change the signs of the off-diagonal elements (6 and 6), and divide the resulting matrix by the determinant:

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

Finally, simplifying each element:

$$(X^T X)^{-1} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix}.$$

(c) Next, we calculate $X^T Y$:

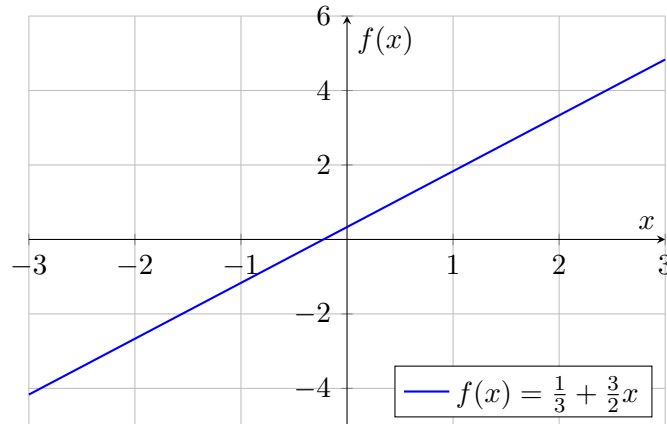
$$X^T Y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 + 3 + 5 \\ 2(1) + 3(2) + 5(3) \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}.$$

Now, multiplying $(X^T X)^{-1}$ by $X^T Y$:

$$\hat{\beta} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix} = \begin{bmatrix} \frac{70}{3} - 23 \\ -10 + \frac{23}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}.$$

Thus, the estimated parameters are $\hat{\beta}_0 = \frac{1}{3}$ and $\hat{\beta}_1 = \frac{3}{2} = 1.5$. The linear prediction function is:

$$f(x) = \frac{1}{3} + \frac{3}{2}x.$$



(d) The predicted vector $\hat{Y}$ is:

$$\hat{Y} = X \hat{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{11}{6} \\ \frac{10}{3} \\ \frac{29}{6} \end{bmatrix}.$$

The residual vector $e = Y - \hat{Y}$ is:

$$e = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} - \begin{bmatrix} \frac{11}{6} \\ \frac{10}{3} \\ \frac{29}{6} \end{bmatrix} = \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix}.$$

Problem 2: Properties of the Hat Matrix & Projection Geometry

In ordinary linear regression, the vector of predicted values can be written as $\hat{Y} = HY$, where $H = X(X^T X)^{-1} X^T$ is known as the *Hat Matrix*.

- (a) Prove that the hat matrix H is symmetric, i.e., $H^T = H$.
- (b) Prove that the hat matrix H is idempotent, i.e., $H^2 = H$. Explain the geometric intuition of idempotency in terms of orthogonal projection onto the column space of X .
- (c) Show that the residual vector $e = Y - \hat{Y}$ is orthogonal to every column of the design matrix X , i.e., $X^T e = \mathbf{0}$.

$$\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i.$$

Solution

- (a) **Showing that H is symmetric**

Recall that the hat matrix is defined as

$$H = X(X^T X)^{-1} X^T.$$

A matrix is called *symmetric* if its transpose is equal to itself, i.e.,

$$H^T = H.$$

We therefore take the transpose of H . To do this, we use the following property of matrix transposes:

$$(AB)^T = B^T A^T.$$

In other words, when taking the transpose of a product of matrices, the order of multiplication is reversed and each individual matrix is transposed. Extending this to three matrices gives:

$$(ABC)^T = C^T B^T A^T.$$

Therefore,

$$\begin{aligned} H^T &= (X(X^T X)^{-1} X^T)^T \\ &= (X^T)^T ((X^T X)^{-1})^T X^T. \end{aligned}$$

Now, $(X^T)^T = X$, since taking the transpose twice returns the original matrix.

Next, we need to show that

$$((X^T X)^{-1})^T = (X^T X)^{-1}.$$

First observe that $X^T X$ is symmetric:

$$(X^T X)^T = X^T (X^T)^T = X^T X.$$

Since $X^T X$ is symmetric, its inverse is also symmetric. Hence,

$$((X^T X)^{-1})^T = (X^T X)^{-1}.$$

Substituting these results back:

$$\begin{aligned} H^T &= X(X^T X)^{-1} X^T \\ &= H. \end{aligned}$$

Thus,

$$\boxed{H^T = H}.$$

Therefore, the hat matrix H is **symmetric**.

(b) **Showing that H is idempotent**

A matrix H is called *idempotent* if multiplying it by itself gives the same matrix:

$$H^2 = H.$$

Starting from the definition of H :

$$H = X(X^T X)^{-1} X^T.$$

Therefore,

$$\begin{aligned} H^2 &= H \cdot H \\ &= (X(X^T X)^{-1} X^T) (X(X^T X)^{-1} X^T). \end{aligned}$$

Using the associativity of matrix multiplication, we can regroup the matrices without changing their order:

$$H^2 = X(X^T X)^{-1} \underbrace{X^T X}_{\text{middle terms}} (X^T X)^{-1} X^T.$$

Notice that the two middle matrices combine to form $X^T X$. Thus,

$$H^2 = X(X^T X)^{-1} (X^T X) (X^T X)^{-1} X^T.$$

We now use the inverse property:

$$A^{-1} A = A A^{-1} = I,$$

where I is the identity matrix. Therefore,

$$(X^T X)^{-1}(X^T X) = I.$$

Hence,

$$\begin{aligned} H^2 &= X I (X^T X)^{-1} X^T \\ &= X (X^T X)^{-1} X^T \\ &= H. \end{aligned}$$

Thus,

$$\boxed{H^2 = H}.$$

Therefore, H is **idempotent**.

Geometric Intuition:

The hat matrix H acts as a projection matrix. It projects any vector Y onto the column space of X , denoted by $\text{Col}(X)$:

$$\hat{Y} = HY.$$

The vector $\hat{Y}$ is therefore the projection of Y onto $\text{Col}(X)$ and already lies inside this space. If we project $\hat{Y}$ again, nothing changes because it is already in the space onto which we are projecting:

$$H\hat{Y} = \hat{Y}.$$

Since $\hat{Y} = HY$, we have

$$H(HY) = HY,$$

or equivalently,

$$H^2 Y = HY.$$

This is precisely the geometric meaning of the idempotent property $H^2 = H$: *once a vector has been projected onto a subspace, projecting it again leaves it unchanged.*

(c) Showing that the residual vector is orthogonal to every column of X

The predicted vector in linear regression can be written using the hat matrix H as

$$\hat{Y} = HY,$$

where

$$H = X(X^T X)^{-1} X^T.$$

The residual vector is the difference between the actual target vector Y and its predicted value $\hat{Y}$:

$$e = Y - \hat{Y}.$$

Since $\hat{Y} = HY$, we can write

$$e = Y - HY = (I - H)Y.$$

To show that the residual vector e is orthogonal to every column of X , we need to show that

$$X^T e = \mathbf{0}.$$

To understand why this condition represents orthogonality, suppose the columns of X are

$$X = \begin{bmatrix} | & | & \cdots & | \\ x_1 & x_2 & \cdots & x_p \\ | & | & \cdots & | \end{bmatrix}.$$

Then,

$$X^T e = \begin{bmatrix} x_1^T e \\ x_2^T e \\ \vdots \\ x_p^T e \end{bmatrix}.$$

Each entry $x_i^T e$ is the dot product between the residual vector e and the corresponding column x_i of X .

Recall that two vectors are orthogonal if their dot product is zero. Therefore,

$$X^T e = \mathbf{0}$$

means that

$$x_1^T e = x_2^T e = \cdots = x_p^T e = 0.$$

Hence, e is orthogonal to every column of X .

Now, using

$$e = (I - H)Y,$$

we have

$$X^T e = X^T (I - H)Y.$$

Expanding the expression:

$$X^T e = (X^T - X^T H)Y.$$

Using the definition of the hat matrix,

$$H = X(X^T X)^{-1} X^T,$$

we get

$$\begin{aligned} X^T H &= X^T X (X^T X)^{-1} X^T \\ &= I X^T \\ &= X^T. \end{aligned}$$

Therefore,

$$\begin{aligned} X^T e &= (X^T - X^T)Y \\ &= \mathbf{0}. \end{aligned}$$

Thus,

$$\boxed{X^T e = \mathbf{0}}.$$

Therefore, the residual vector e is orthogonal to every column of X .

Geometric Interpretation:

The hat matrix H projects Y onto the column space of X :

$$\hat{Y} = HY.$$

Therefore, the residual

$$e = Y - \hat{Y}$$

is the part of Y that is left after removing its projection onto $\text{Col}(X)$.

The projection $\hat{Y}$ lies inside $\text{Col}(X)$, while the residual e is perpendicular to this entire space. Since every column of X lies in $\text{Col}(X)$, e must be perpendicular to every column of X .

This is expressed algebraically by

$$\boxed{X^T e = \mathbf{0}}.$$